

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Classified Practice

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every source Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Classified Practice

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.

Q007 Written**Find the four quadratics satisfying the four source triples of derivative and function conditions.**

Classified Practice

Differentiation and Tangent Lines

Q008 Written

Find the quadratic $(f(x))$ satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Q009 Written**Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.**

Classified Practice

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three source functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

Q011 Written

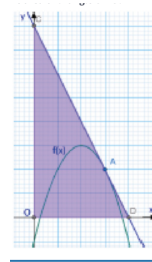
Find the tangent line for all nine source Exercise functions at their given points.

Classified Practice

Differentiation and Tangent Lines

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice

Differentiation and Tangent Lines

Q013 Written

Solve the two source Warm Up demands: tangents from $(C(0,2))$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 Written

Solve the two source Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Classified Practice

Differentiation and Tangent Lines

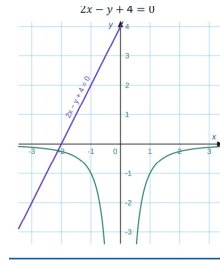
Q015 **Written****Solve all six source Exercise demands involving external points or prescribed gradients.**

Classified Practice

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Choose the correct option.**

A x^{n-1}

B nx^{n-1}

C $n + x$

D x^n

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Choose the correct option.

A πh

B $2\pi r h$

C $2\pi r$

D πr^2

Classified Practice

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:**Choose the correct option.**

A -6

B -3

C 0

D 3

Q020 MCQ**The challenge function can be written as:****Choose the correct option.**

A $x^{5/3} + x^{-2}$

B $x^5 + x^2$

C $x^{3/5} + x^2$

D $x^{5/3} + x^2$

Classified Practice

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 MCQ

The condition $f'(0) = -2$ gives:**Choose the correct option.**

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Choose the correct option.**

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ

The challenge coefficient a is:**Choose the correct option.**

A 1

B 2

C 3

D 6

Classified Practice

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Choose the correct option.**A $f(a)$ B a C $f'(a)$ D $1/f'(a)$

Q026 MCQ**The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:****Choose the correct option.**

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

Classified Practice

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q028 MCQ**The area of triangle (ODC) in the challenge is:****Choose the correct option.****A** 8**B** 12**C** 16**D** 32

Classified Practice

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Choose the correct option.**A $((0,0))$ B $((a,f(a)))$ C $((f(a),a))$ D $((1,f'(a)))$

Q030 MCQ**The gradient of $2x - y + 4 = 0$ is:****Choose the correct option.****A** -2**B** -1**C** 1**D** 2

Classified Practice

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:**Choose the correct option.**

A $-2/x^3$

B $2/x^3$

C $1/x^2$

D $-1/x^3$

Q032 MCQ**The parallel tangent in the challenge is:****Choose the correct option.**

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1$, $f'(1) = 3$, $f(1) = 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 Concept Quiz · MCQ

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

- A three function values
- B two derivatives and one function value
- C one derivative only
- D the tangent gradient only

Classified Practice

Differentiation and Tangent Lines

Quiz MCQ 7 **Concept Quiz · MCQ****The tangent equation in point-slope form is:****Choose the correct option.**

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 Concept Quiz · MCQ

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B $1/3$

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C